

Creating Infinite Characters From a Single Template: How Automation May Give Super Powers to 3D Artists

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Figure 1: On each row, the leftmost image shows a template character and the next images show a few automatically generated variations. Top: template supplied courtesy of Avalon, bottom: Genshin Daz3d template.

ABSTRACT

Game character creation is a time-consuming and highly manual process that often requires several days to complete one non-player character. In addition, each game has unique character art style and technical specifications (mesh, rig, textures). This explains why game worlds with thousands of unique characters are rare. To address this problem, we propose an automated character generation pipeline that can produce a full-body, rigged, dressed, and accessorized 3D character in less than a minute from an initial template character, based on 2D pictures and/or facial text descriptors. This pipeline helps scale up character creation and populate games with unlimited variations at reduced cost, while ensuring consistency with the game art style.

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1 INTRODUCTION

Populating games with varied characters is challenging because character design usually involves extensive manual work by skilled 3D artists. Moreover, this requires the adoption of a unique game aesthetic by several artists, which may involve multiple development iterations. Owing to their impact on budget and development time, these challenges may keep game designers from achieving their vision, overarching narrative and intended player journey.

Existing tools to support character creation either demand extensive manual interaction, as seen in traditional character creator software, or lack customization and adaptability to predefined technical specifications, as observed in the MetaHuman creator [Fang

et al. 2021] or in recent AI-driven 3D character generation methods [Altundaş and Karaarslan 2023; Hong et al. 2022; Liao et al. 2023].

This work investigates how automation can help solve these problems by combining modern deep learning and more traditional computational geometry methods to streamline non-creative, painstaking tasks involved in character creation, while allowing artists to focus on genuinely creative work. Our pipeline leverages a single initial full-body template character to quickly create multiple “child” characters based on user-supplied photos, 2D concept art pictures, or text descriptions of facial traits. Since it does not require special art skills, it may be used by both game developers and players through an in-game character editor.

2 PIPELINE

The proposed character generation pipeline is sketched in Figure 2 and the different steps comprising it are outlined below.

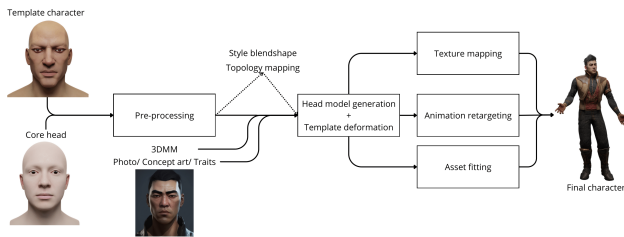


Figure 2: Pipeline overview: from user input (2D picture or text description) to an animatable 3D character.

Pre-processing stage: shrink wrapping and style extraction. To register a certain art style in our system, we need to wrap a template character provided by the user around a fixed-topology head shape template, hereafter referred to as *core head mesh*, which is chosen as the mean shape of a 3D morphable model (3DMM) representing a broad range of realistic human head shape variations [Dias et al. 2022]. This process is carried out automatically using an in-house implementation of the iterative closest point (ICP) algorithm, which requires 10–15 minutes on a standard CPU regardless of the template mesh topology. It enables us to convert the 3DMM to the user’s topology, and to further project the template head shape onto the 3DMM, i.e., find the closest 3DMM-generated human head shape to the template head. The difference between the template and its projection then defines the *stylization* delta blendshape, which encodes the non-human aspects of the user style (cartoony, “chad-looking”, ...). Note that the projection step may be skipped if the user provides both a human template and a stylized one.

Human head model generation. The creation of a new character starts with the generation of a head mesh and albedo maps (for head, eyes and hair) in the core mesh topology. This is achieved using an extension of our avatar creation algorithm [Dias et al. 2022] that can work either deterministically from a picture (frontal photo, human-looking 2D concept art), or randomly from a set of demographic and morphometric descriptors (age, gender, ethnicity, facial feature lengths, angles and areas). Using the point correspondences determined in the initial ICP stage, we compute a thin-plate spline

(TPS) model to deform the template head into the newly created head shape. This TPS is regularized locally to satisfy pre-defined local stiffness constraints, e.g., eyeball / teeth rigidity, and subsequently applied to the whole template shape. The model is further customized by linear application of the stylization blendshape and other optional blendshapes, e.g., for body variations.

Texture mapping. The albedo maps are transferred to the template UV space by means of sparse 2D interpolation, then optionally submitted to low-pass image filtering and color remapping depending on the desired style (for instance, anime characters may undergo strong low-pass filtering to uniformize the color spatially and to produce colorful skin, eye and hair tones).

Animation retargeting. Each bone in the bind pose is first repositioned and re-oriented via a rigid transformation that locally fits the vertex displacements from the template to the new character. The resulting bone transformations are further processed to enforce bone coplanarity and pairwise orientation constraints. Animations are then retargeted using conventional inverse kinematics solvers.

Asset fitting. Existing attachments (hair, outfits, accessories) designed for the template are transferred to the new character by interpolating the 3D vertex displacements from the template. Similarly to the template deformation step, the interpolated displacements are regularized locally to enforce user-specified rigidity constraints while maintaining overall spatial smoothness. For instance, eye-glass rims and temples may be constrained to remain rigid while rotating with respect to one another; shoe soles may remain rigid while the tongues and collars may deform.

3 USE CASE

The proposed character generation pipeline is integrated in a commercial game development support software called Popul8, which is being used for several MMOs under development. It was used by Colossal Order for Cities: Skylines II, a AAA game released in 2023, where 30 base characters, each with 146 animations and 102 attachments, were created from two templates (male and female) using the proposed pipeline, and a total of 2 600 unique non-player characters were created by blending between the base characters and templates. Importantly, all characters were created within a one-month time frame, taking into account several fine-tuning iterations, something that would have normally taken 4+ months to accomplish through traditional methods.

It is estimated that this capability saved asset budget costs by as much as 50%. The methodology allows for quick updates in that, at any point in time, new characters and new assets can be added, and all assets are automatically fitted to all characters.

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